BLUEPRINT BIOSECURITY · NOVEMBER 9, 2026

Help more people find respirators that fit

Extend MasqFit to Android and standard cameras, and test facial features tailored to different respirator styles.

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Draft for discussion | Android and web research | Open research outputs

← Short concept note

> **Research question**
>
> Can a guided RGB capture plus one inexpensive physical scale reference recover facial measurements accurately enough to support useful respirator recommendations across Android phones and web browsers?
>
> **Request:** $196,800 total; $66,800 initial option. Twelve active research months, approximately fourteen calendar months if the award spans planned parental leave.

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1. Public-health purpose and existing work

People often need to try multiple respirator models and sizes before finding a suitable fit. MasqFit aims to make that search more informed by combining facial geometry with observed fit evidence. Its existing iOS app uses Apple's TrueDepth/ARKit workflow; the proposed work would investigate an ordinary-camera alternative and independently test its practical usefulness.

Kanro's $50,000 award in 2024 supported the initial research and app. Edderic led participant recruitment, measurement collection, modeling, and the released product. At CDC/NIOSH NPPTL's recommendation, he has been invited as an external expert to present MasqFit at a NASEM session on November 17, 2026, sharing lessons from its research and development that could help improve NIOSH's own upcoming virtual fit-testing apps.

Blueprint's respiratory protection program addresses barriers to effective respirator use. This project could contribute an open measurement method and evidence about selection guidance, including failure conditions. It is an exploratory submission following Victoria's invitation, rather than an application to a listed PPE RFP. Sources 1–3.

The camera workflow is an experimental prototype, not a validated Android product. Geometry estimates do not measure the worn respirator's seal; fit testing remains the relevant endpoint for the proposed human study.

The existing flow: iPhone capture → measurements → mask ranking

MasqFit already connects a face scan to a personalized shortlist of respirators. These actual screenshots from the NASEM presentation show the existing iOS capture and measurement screens, followed by the ranking in Breathesafe.

1 · Capture facial geometry

Use a compatible Face ID-capable iPhone with a TrueDepth camera and ARKit to capture facial geometry.

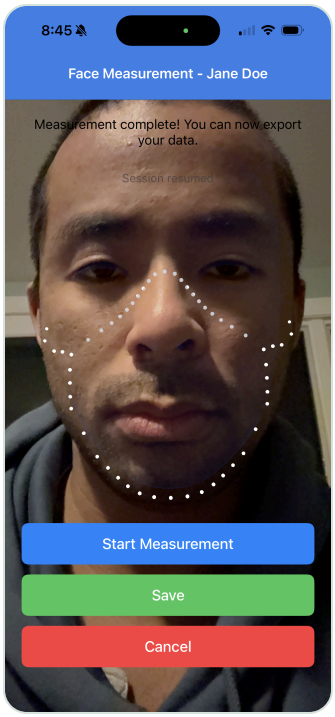
2 · Calculate measurements

Compute the selected facial paths in millimeters and save the measurements.

3 · Rank respirators

Use those measurements with the recommender to order masks by predicted fit score in Breathesafe.

1 · Facial capture on iPhone



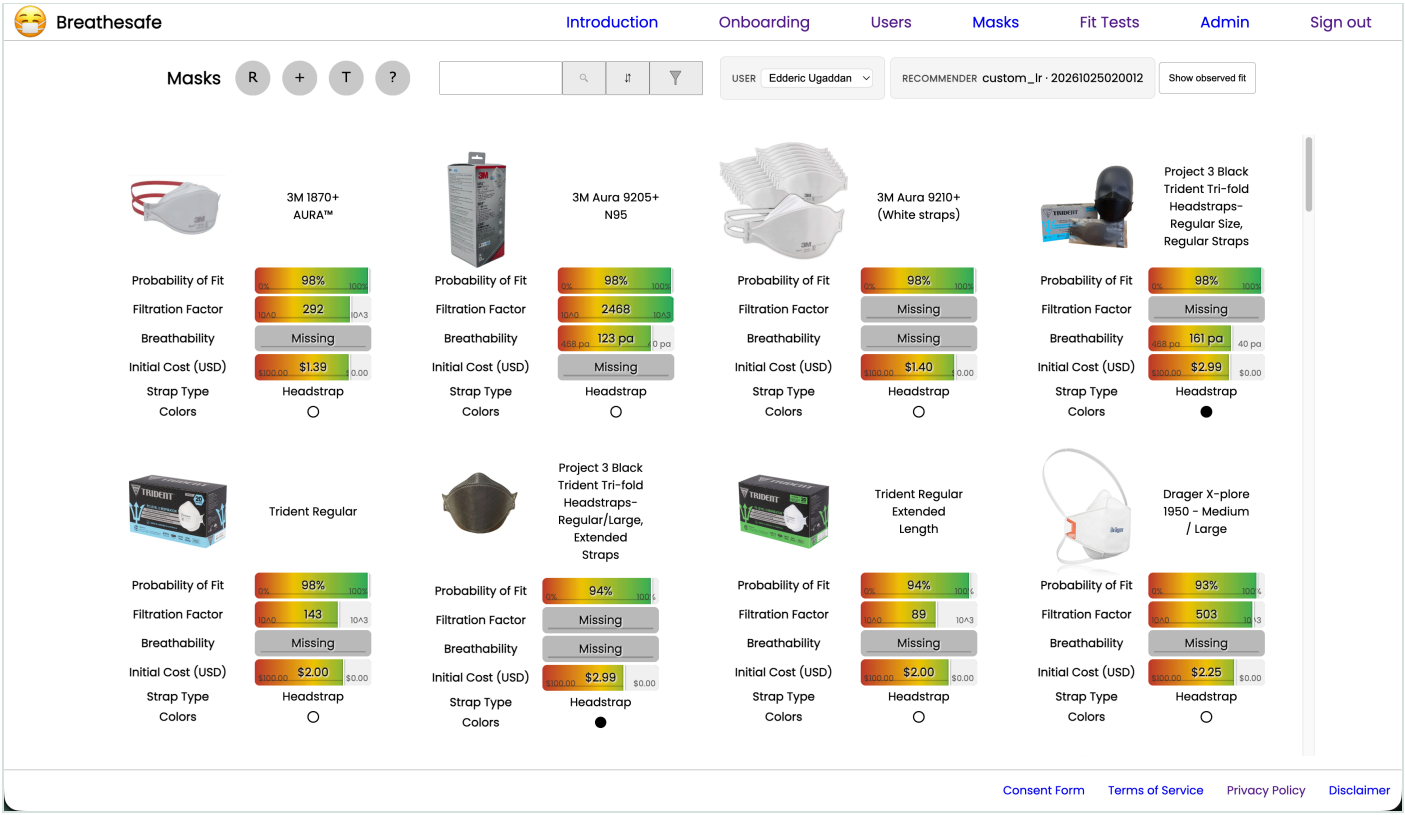
Selected measurement points are overlaid on the face during the TrueDepth/ARKit workflow.

2 · Measurements in millimeters



The app displays five path aggregates: nose, strap, top cheek, mid cheek, and chin. Nose, top cheek, mid cheek, and chin form the current model's perimeter proxy; the collected strap path does not directly enter this model.

3 · A personalized mask ranking in Breathesafe



Actual October 29, 2026 screenshot using the October 25 custom_lr model, with Observed Fit hidden. This is an example of the ranking interface, not the November 8 evaluation snapshot. The displayed "Probability of Fit" percentages are model scores; their calibration as individual success probabilities has not been established. They are separate from the held-out top-1/top-3 hit rates explained later.

Android and standard-camera support would extend this same useful sequence—capture, measure, and rank—to people without the current iPhone depth hardware. The proposed research will validate that alternative measurement source and test whether style-specific features improve the ranking.

Source: presenter-supplied screenshots preserved in the November 17 NASEM presentation assets. Screenshots illustrate the implemented workflow; they do not independently validate measurement accuracy or respirator fit.

Expand access beyond iPhone depth hardware

> **Roughly 3–4× potential global smartphone reach, with additional browser access.**
>
> Android support and standard-camera capture would let more people use equipment they already own. The opportunity is broader than porting an app: it removes the current scan-based workflow's dependency on compatible iPhone TrueDepth hardware.

Two measures of potential access; different denominators

Source	Observed platform base	Implication for access

Source	Observed platform base	Implication for access
Statcounter · October 2026	Android 72.36%; iOS 27.61% of global mobile web usage.	(72.36 + 27.61) / 27.61 ≈ **3.62×** the iOS-only usage base when both are included.
Counterpoint · 2025 active devices	Approximately one in four active smartphones is an iPhone.	The entire active smartphone base is approximately **4×** the iPhone base. This describes non-iPhone devices collectively, not an independently measured Android-user count.

The multiplier describes the **total potential base**, not 3–4 times as many additional users. Usage shares and installed devices are different measures, neither of which counts unique MasqFit users. They support an access opportunity; they do not establish a guaranteed minimum expansion or forecast adoption. The currently compatible TrueDepth iPhone base is itself a subset of all iPhones.

A standard RGB-camera workflow could also accommodate supported laptop/desktop webcams and phones without the current depth hardware. No extra numeric audience is assigned to those channels. A tape measure provides inexpensive physical scale; guided capture and calibration must still be usable and accurate. Server processing, upload access, device quality, and local respirator availability constrain realized reach. Phase C/E will report supported devices, completion, time, processing cost, and failures rather than assume universal compatibility.

Sources: Statcounter Global Stats, October 2026; Counterpoint Research, February 10, 2026, reporting the 2025 active installed base. Consulted November 9, 2026. The arithmetic above is a project inference from those sources.

2. The current camera-to-measurement pipeline

1 · Capture

Multiple ordinary-camera views; retain face and upper-neck regions with explicit coverage checks.

2 · Reconstruct

Photogrammetric surface and fixed COLMAP cameras; triangulate candidate correspondences across views.

3 · Smooth

Gaussian convolution of a supported 3D implicit field; extract a candidate surface.

4 · Scale and check

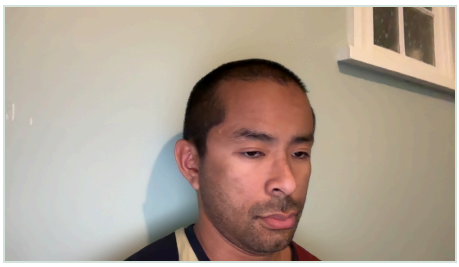
Use a tape-measured arc for millimeters; compare other dimensions with physical readings.

Example input photos from one photogrammetry run

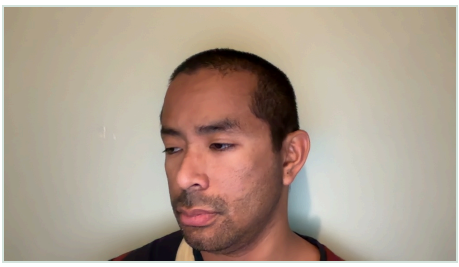
The reconstruction starts with overlapping ordinary-camera views of the same face. These six frames are sampled from the 61-image sequence used for the smoothed surface and experimental paths shown below.



View 1 · front



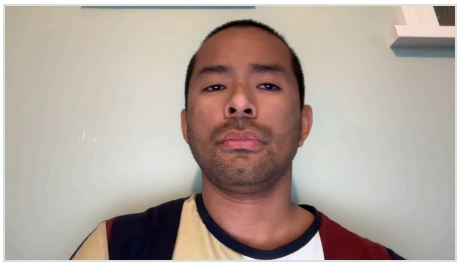
View 2 · turn_11



View 3 · turn_22



View 4 · turn_35



View 5 · turn_48



View 6 · turn_55

Actual source-photo previews from one capture sequence; unannotated views, without the measurement-path overlays. They show variation in viewing angle and visible facial regions. The experiment used all 61 source images—not just these six—and this figure does not establish the minimum number of views needed for a reliable result. These are views of one person, not independent participants or validation captures.

Photogrammetry and landmark support

The current illustrated experiment uses 61 images from one source capture sequence. The upper-neck reconstruction reuses those images with expanded masks; it is not a second independent capture. Dense learned image correspondences suggest candidate points, while triangulation uses the original photogrammetry cameras. FLAME supplies correspondence identities and candidate regions, rather than fitted facial coordinates or metric scale.

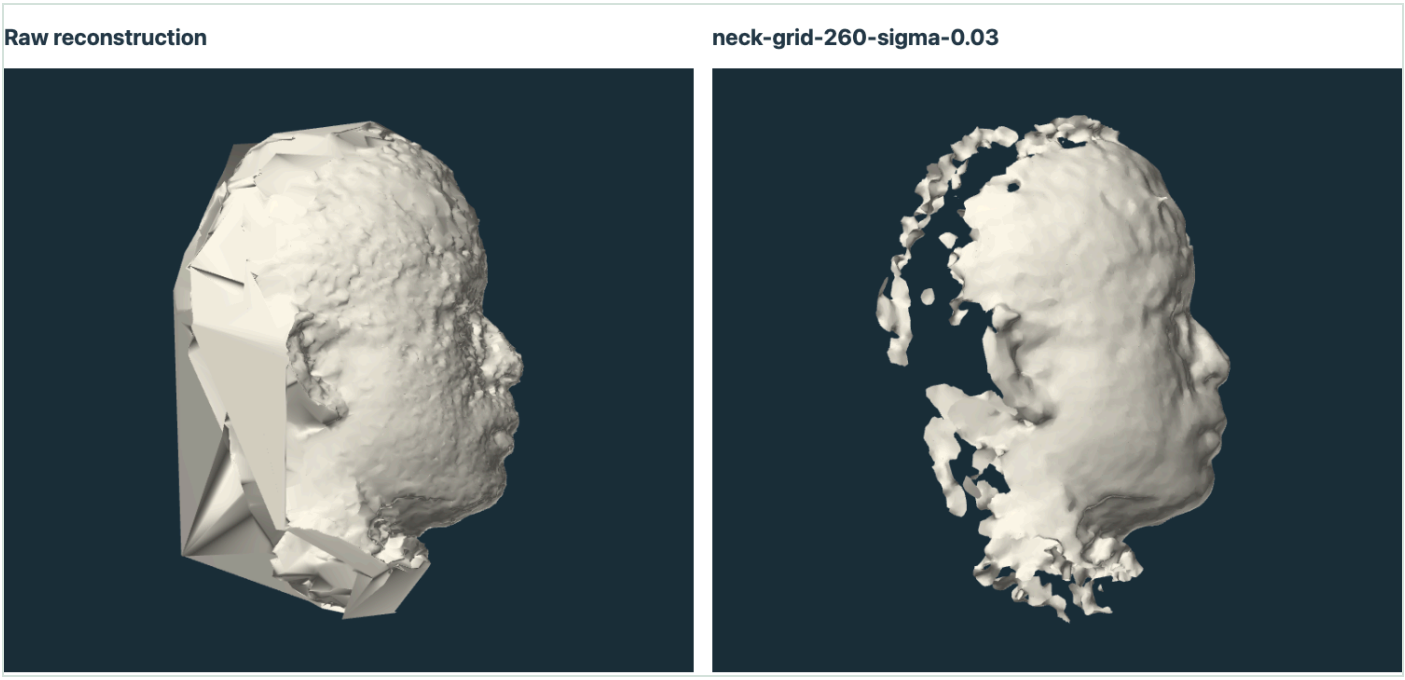
Exploratory support checks require at least three agreeing visible views, six-pixel reprojection tolerance, at least three degrees of parallax, and proximity to eligible raw geometry. Crop, mask, occlusion, and triangle-quality checks reject unsupported candidates. These checks detect disagreement; correlated detector bias can still pass them.

Smoothing by 3D convolution

The mesh is not watertight, so the experiment builds a local oriented point-to-plane field from quality-filtered surface samples rather than assuming a globally valid signed distance. Gaussian convolution uses normalized weights to keep unknown voxels from acting like measured zeros:

```
smoothed field = Gσ(field × support weight) / Gσ(support weight)
```

Marching cubes extracts a candidate surface within the retained support band. The selected upper-neck candidate uses σ = 0.03 of the reference eye-proxy span, approximately 1.40 voxels, on a 180 × 260 × 191 grid. The eye proxy normalizes experimental settings; it is not the physical calibration measurement. Zero-filter remeshing is retained to separate remeshing effects from filtering.



Actual pilot reconstruction: raw surface at left, selected Gaussian candidate at right. Shared-view comparison illustrates smoothing and removal of some poor geometry. It does not establish improved anatomical accuracy; missing regions, inferred surfaces, and source bias remain.

Smoothing can move landmarks, join nearby surfaces, or alter the nose. New geometry does not become supported merely because it looks smooth. Candidate points must pass raw-surface checks before being assessed on the smoothed surface. Complete lengths remain unavailable when controls or connected paths are missing.

From reconstruction units to millimeters

The calibration is the bitragion–subnasale arc: left tragion → subnasale, at the base of the nose → right tragion. A flexible tape supplies physical length. In this experiment the three readings were 288, 280, and 284 mm, with a median of 284 mm. The corresponding model surface route measured 4.28220 reconstruction units.

```
scale = 284 mm / 4.28220 units = 66.32106 mm per unit
estimated dimension in mm = reconstructed dimension × scale
```

The same scale is held fixed for nose width, face width, face length, and the perimeter check. Their readings do not recalibrate it. The calibration arc cannot count as a successful validation endpoint because its length defines scale. The 280–288 mm range alone implies approximately ±1.41% variation around the median scale; this sensitivity is not a confidence interval. Tape tension, soft-tissue contact, anatomical placement, and bridging of concavities can introduce additional bias.

3. Preliminary measurement results

0.46%	1.37%	1.98%
Nose-width absolute relative error	Face-width absolute relative error	Face-length absolute relative error

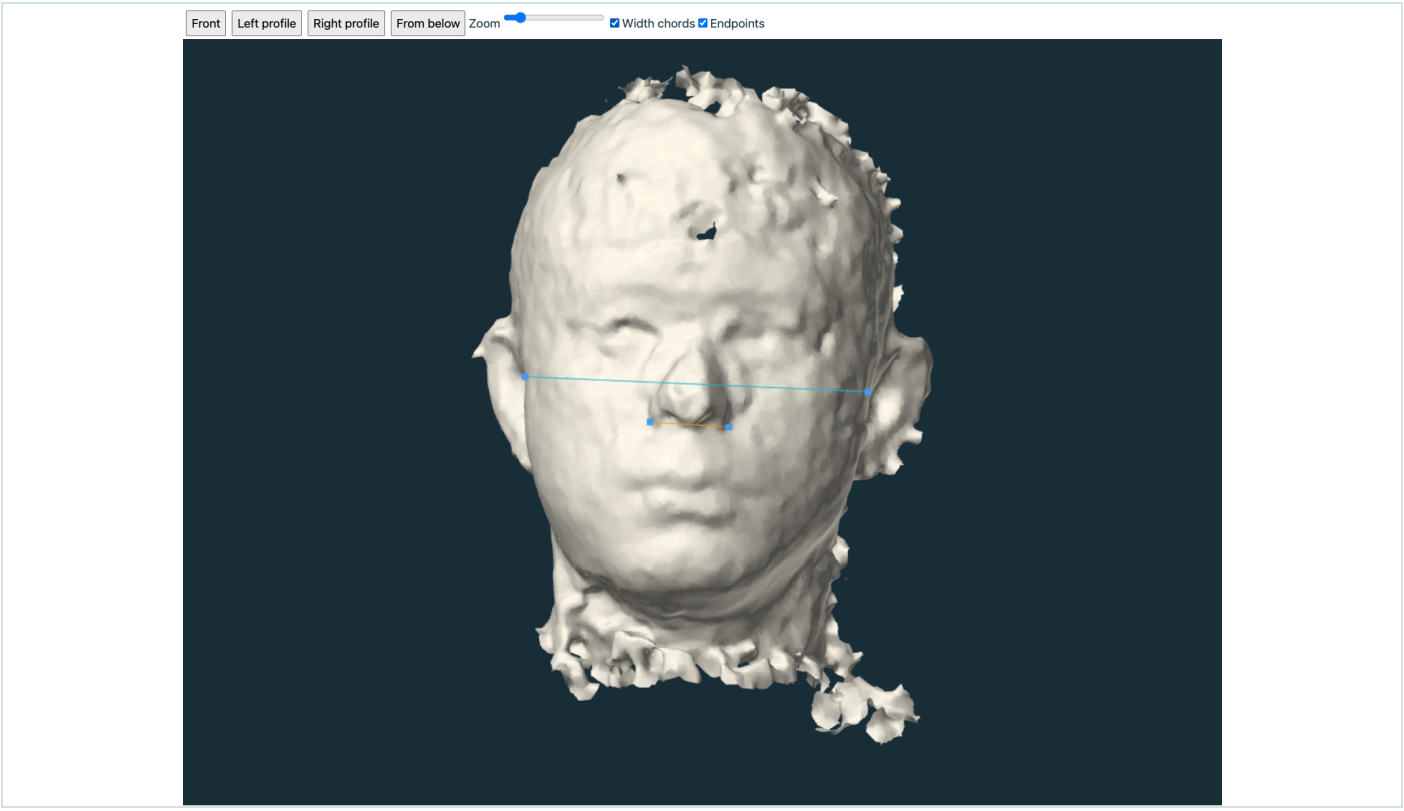
Exploratory comparison from one source capture sequence, scaled with a separate tape arc. Endpoint review is pending; these are not population accuracy estimates.

One-source-capture physical comparisons · all lengths in mm

Dimension	Physical readings	Reference median	Estimate	Signed error	Absolute relative error
Nose width · proposed alare contacts	35, 36, 35	35.00	35.16	+0.16	**0.46%**
Face width · proposed zygomatic contacts	151.5, 150.9, 150.7	150.90	152.97	+2.07	**1.37%**
Face length · sellion–menton chord	115.5, 114.6, 113.9	114.60	112.33	-2.27	**1.98%**

```
signed error = estimate − reference median
absolute relative percent error = 100 × |estimate − reference median| / reference median
```

Nose width and face width are straight 3D chords between proposed soft-tissue contact candidates. Their endpoint pairs maximize horizontal breadth within fixed correspondence regions while restricting height mismatch. Face length uses the provisional sellion–menton chord. These definitions still require anatomical review and matching to physical caliper contacts.



Proposed width contacts on the pilot surface. Colored chords are measurement candidates, not independently accepted anatomical landmarks. Their source photographs remain in local research records and are not embedded in this proposal.

> **What the small errors do—and do not—show**
>
> These are encouraging feasibility comparisons after iterative work on the same source capture, not results from a locked method on unseen people. Repeated physical readings are not independent participants. All three outputs remain provisional, with accepted-measurement fields unset in the source reports.
>
> The current selection function excludes physical width readings, but candidate-region and endpoint rules were revised during development. An earlier width version gave approximately 4.50% nose-width error; the current version gives 0.46%. We therefore cannot treat the current result as blind validation or use it to promise sub-percent accuracy.

Evidence provenance

The values above come from the November 9 width-endpoint report (version 4), the matching face-length check, the fixed arc-calibration report, and the selected upper-neck Gaussian experiment. Their source and selected-mesh hashes agree. A separate local evidence register preserves exact values, report hashes, and input paths for audit; it is not a participant-data release.

Improve the facial representation for each respirator style

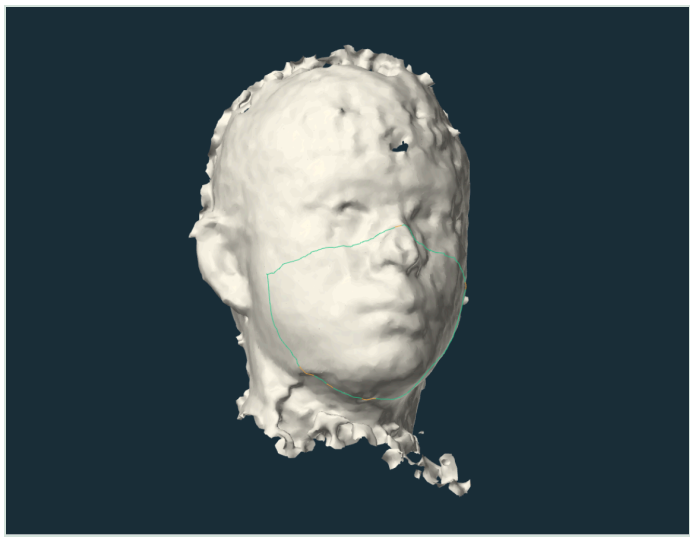
This is a second advance alongside cross-platform access. The current iOS recommender uses a facial perimeter proxy, mask perimeter, family/style terms, and a strap contribution. Its nose path contributes to the aggregate but is not independently represented as bridge height or lateral nasal slope. Two faces with the same aggregate can therefore receive the same geometry-dependent score for a given mask despite different local shape.

We will test whether preserving local features—and allowing their importance to vary by respirator style or supported product family—adds predictive value. The rows below are proposed hypotheses, not established seal mechanics or measured improvements.

Two existing experimental paths on the same 3D surface

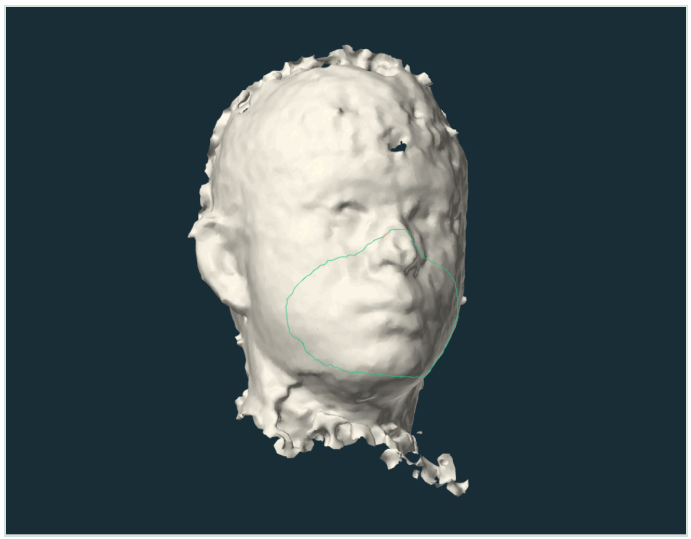
These are the saved duckbill/trifold and bifold contact-route candidates, projected onto the same Gaussian-smoothed photogrammetric surface. The side-by-side views use identical orientation and scale, making their proposed cheek and chin contours directly comparable.

Duckbill / trifold candidate

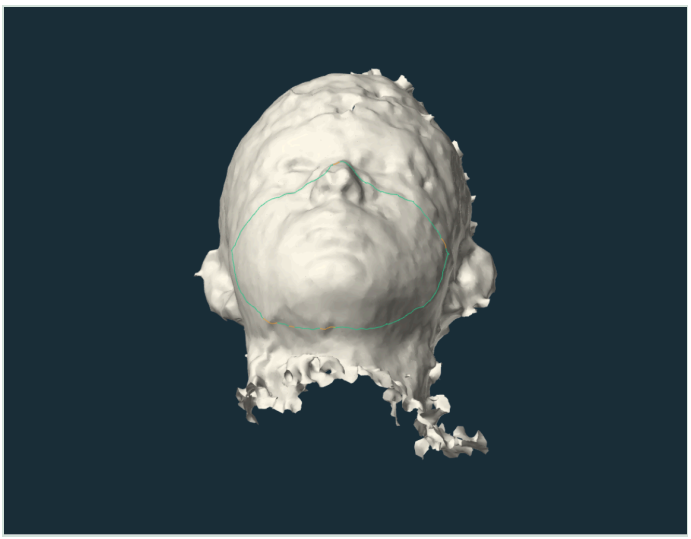


Shared experimental route for the duckbill/trifold use case.

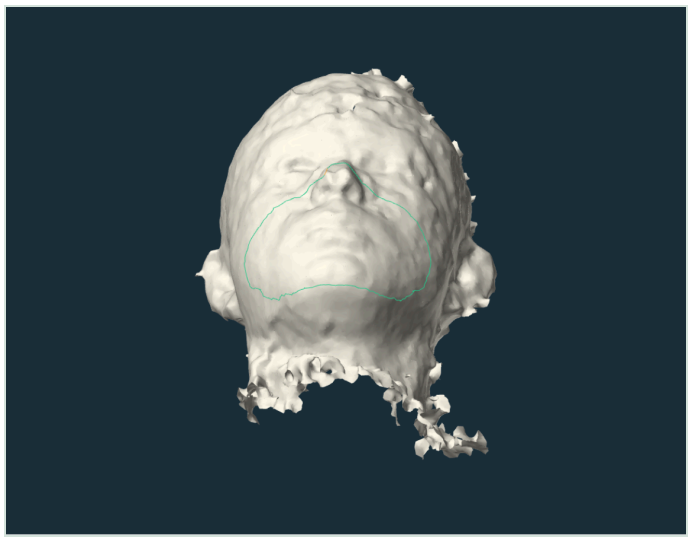
Bifold candidate



Separate experimental bifold route on the same facial reconstruction.



Duckbill / trifold · lower view.



Bifold · matching lower view.

Green segments pass the experiment's sampled source-surface/photo consistency checks; orange segments have incomplete support. The overlay can show through the surface. These are geometry-derived route proposals awaiting path/anatomical review, not verified respirator seals or accepted physical measurements. Different products within a style may require different boundaries. The figures motivate testing style-specific geometry; they do not establish which path predicts fit better.

Candidate facial features to investigate by respirator style

Style	Candidate local geometry	Question to test
Duckbill	Lower-cheek/jaw span, chin projection and contour, nose-to-chin span, and nasal/upper-cheek shape.	Do separate lower-face and upper-seal features distinguish fit beyond a single total perimeter?
Trifold / boat	Upper-cheek breadth, bridge height/protrusion and lateral nasal slope, lower-panel/chin contour, and nose-to-chin span.	Does representing upper and lower panel regions separately improve ranking for models with different panel dimensions?
Bifold	Nasal root/bridge geometry, alar width, lateral nasal slope, cheek breadth, and nose-to-chin span.	Do local nasal and cheek features add useful information within a style whose models and sizes can share similar aggregate dimensions?
Cup	Face width and nose-to-chin span, nasal bridge height/protrusion, cheek contour, and chin projection relative to the product's sealing boundary.	Do local contour and size features better distinguish fit for cup-shaped models than aggregate perimeter alone?
Quarter-mask elastomeric	Product-specific seal footprint, bridge height/protrusion, alar and upper-cheek contour, and local contact widths.	Can geometry matched to the actual product boundary better distinguish sizes or nose-bridge variants than a general face perimeter? An under-chin route will not be assumed appropriate for every product.

Separate better features from better camera measurements

First compare feature sets using independent reference geometry; then test the same definitions on ordinary-camera measurements. Include the current ARKit-to-perimeter workflow as the end-to-end incumbent baseline. Extra nasal, cheek, chin, or ear-adjacent landmarks need their own definitions, visibility checks, repeatability assessment, and validation; existing mesh vertices will not be relabeled as anatomical points without review.

Define each feature's endpoints, reference planes, units, surface-versus-chord interpretation, and relationship to the product's sealing boundary. Distinguish bridge height, protrusion, slopes, and categorical bridge variants. Match facial features with mask design, model/size, and variant identity; a face-only score cannot rank alternative masks for the same person.

Bounded model comparison and evidence required

1. **Baseline:** current perimeter representation with the existing family/style/strap structure.
2. **Local additions:** baseline plus a small prespecified subset of independently validated nose, cheek, or chin features.
3. **Style-dependent candidate:** regularized feature-by-style effects, with family/variant refinements only where supported by participant and product coverage. Use shared/shrunk effects when data are sparse instead of fitting a separate unconstrained model for every style.

Keep participants, tested options, label definitions, and tuning budgets comparable. Feature selection, normalization, imputation, and tuning stay within participant-grouped development folds. Report paired changes in top-1/top-3 ranking relevance with the same conditional eligibility definition, discrimination/calibration where estimable, and measurement/capture burden. Report eligible people and candidate observations by style, with participant-level uncertainty; do not compare styles on incompatible candidate sets or interpret sparse cells as evidence of superiority.

The initial comparison fits within the ten PI days ($6,000) already reserved in B for feature experiments, with reference collection in A/C. Feasibility will prioritize a small number of features and styles with adequate linked outcomes. This allocation does not promise five independently powered style-specific studies. The prospective pilot retains its bounded FFR scope and up-to-seven-option union. Quarter-mask elastomeric work uses suitable consented feature-linked records when available; otherwise document the evidence gap and scope a follow-up rather than claim validation.

Freeze one selected recommender before the prospective pilot. If local features or style interactions do not reliably improve results, retain the existing model. Deliver a versioned feature/landmark specification, reproducible comparison, style-level evidence summary, and model-selection decision.

Existing iOS recommender: what top-1 and top-3 mean

The live Breathesafe dashboard, checked November 9, 2026, reports the following evaluation results for the November 8, 2026 recommender model used with the iOS measurement workflow:

Retrospective ranking relevance · five user-grouped folds

Metric	Reported fold mean	What counts as a hit for an eligible user?
Top-1	51.7%	The highest-scored held-out test observation has a positive fit label.
Top-3	82.2%	At least one of the first three ranked held-out test observations has a positive fit label.

In each held-out user group, the model ranks that person's recorded test observations by predicted fit score. The metric includes only users with at least one positive held-out label. A hit is scored once per eligible user for each k; the dashboard then reports the unweighted mean of the five fold-level rates. These percentages are not a pooled count out of 100 users.

Worked illustration · all people, counts, and fold rates below are invented

1 · Turn one person's ranking into two hit scores

Example B has at least one positive held-out fit label, so this person is eligible. The model ranks their previously tested observations from highest to lowest predicted score:

RANK 1	RANK 2	RANK 3	RANK 4
Observation A	**Observation B**	**Observation C**	**Observation D**
Negative label	Negative label	Positive label	Negative label

Top 3 includes one positive result

Top-1: miss → 0

Only rank 1 counts. Its label is negative.

Top-3: hit → 1

Ranks 1–3 count. At least one label is positive.

The top-3 score is 1, not 1/3: this asks whether the shortlist contains any positive result.

2 · Average eligible people within one fold

Imagine Fold 1 contains four eligible held-out people. Example B above contributes 0 to top-1 and 1 to top-3:

Illustrative person	Top-1 hit score	Top-3 hit score
A	1	1
B · ranking shown above	0	1
C	0	0
E	1	1
Fold 1 rate	2 / 4 = **50%**	3 / 4 = **75%**

Any person with no positive held-out label is excluded from both denominators. The denominator counts eligible people, not masks, test rows, or every enrolled participant.

3 · Repeat with five held-out person groups

Each row represents a separately trained evaluation model: train on four groups, then evaluate the remaining group. **T** marks the held-out group; gray dashes mark training groups. Every observation from the same person stays in one group.

Invented fold results to demonstrate the averaging rule

Evaluation	Person groups 1–5	Eligible people	Top-1 rate	Top-3 rate
Fold 1	T – – – –	4	2/4 = 50%	3/4 = 75%
Fold 2	– T – – –	5	3/5 = 60%	4/5 = 80%
Fold 3	– – T – –	10	4/10 = 40%	8/10 = 80%
Fold 4	– – – T –	4	1/4 = 25%	4/4 = 100%
Fold 5	– – – – T	5	3/5 = 60%	4/5 = 80%

Illustrative top-1 mean: 47%

`(50% + 60% + 40% + 25% + 60%) / 5`

Illustrative top-3 mean: 83%

`(75% + 80% + 80% + 100% + 80%) / 5`

Each fold has equal weight even though the eligible-person counts differ. Pooling all hit people across folds would give 13/28 ≈ 46.4% for top-1 and 23/28 ≈ 82.1% for top-3 in this toy example; those are different statistics from the mean of fold rates.

Actual reported results: 51.7% top-1 and 82.2% top-3. These come from the November 8 model's five-fold evaluation, checked November 9, 2026. The illustrative 47% and 83% above explain the calculation only; they are not a reconstruction of the actual folds, whose counts and individual rates are unavailable in the dashboard snapshot.

Additional hit/miss examples

The following are invented examples explaining the metric, not additional study participants. "Positive" and "negative" refer to recorded fit labels; rows are ordered from highest to lowest predicted score.

Example	Ranked observations, positions 1 → 4	Top-1	Top-3
A	Positive → negative → negative → negative	Hit	Hit
B	Negative → negative → positive → negative	Miss	Hit
C	Negative → negative → negative → positive	Miss	Miss
D	All recorded held-out observations are negative	Excluded	Excluded

Example B shows the practical motivation for a shortlist: a useful result can appear near the top even when the first choice is unsuccessful. Example C shows that a known positive result may still rank below the top three. Example D explains the conditional denominator: this evaluation does not measure success among people with no recorded positive held-out label.

> **Encouraging ranking evidence; prospective usefulness is the next question.**
>
> The ranked candidates are observed test rows, not the full respirator catalog, and repeated observations can remain. Labels combine formal-test sources and constructed outcomes; this is not a formal-fit-test-only success rate. Fold denominators and confidence intervals are not exposed by the dashboard snapshot.
>
> Consequently, 82.2% does not yet mean that an arbitrary new user has that chance of finding a fitting respirator among three newly recommended models. Nor does it measure all three recommendations fitting, calibrated score probabilities, or Android camera accuracy. The proposed independent geometry benchmark and prospective fit pilot test those next steps separately.

Accuracy, speed, and cost: compare three routes to a useful recommendation

Cost-effectiveness is a research endpoint alongside measurement accuracy. We will compare the planned photogrammetry-based approach with direct fitting of a FLAME 3D model to photos and the existing TrueDepth/ARKit workflow. More detailed reconstruction is worthwhile only if its benefit justifies its capture, latency, infrastructure, and staff burden for the intended setting.

Three alternatives to benchmark on matched participants

Approach	Workflow and potential tradeoff	What is known now / what remains to measure
Photogrammetry + 3D smoothing	Reconstruct a surface from multiple RGB views, smooth it, identify measurement paths/features, and apply a physical scale reference. It may retain useful local shape, but capture, reconstruction, review, and retries can add time and cost.	The current illustrated experiment uses 61 same-sequence images and has preliminary independent-span comparisons. A matched end-to-end timing, per-user cost, and advantage over direct FLAME fitting have not been established.
Direct-photo-to-FLAME fitting	Fit a shared parametric 3D face directly to photo evidence, avoiding a separate photogrammetric reconstruction. It could be simpler or faster; model priors and missing image constraints may limit recovery of individual local shape. A physical scale reference is still required.	The existing baseline uses nine training photos and four additional validation photos; it is not a single-photo feed-forward estimator. One local run recorded 6.96 seconds internally, with physical scale uncalibrated. That timer excludes participant capture, scale collection, uploads/queueing, and delivery of an accepted recommendation; cache/cold-start conditions are not normalized against the other methods.
TrueDepth / ARKit	Use the existing metric facial-capture workflow on a compatible Face ID-capable iPhone. Local capture may be operationally convenient where devices already exist; provisioning devices and supporting access may matter elsewhere.	A working iOS product is available, but matched accuracy, total turnaround time, and fully allocated per-person cost need the same study. Existing iPhones do not make staff time or device cost universally zero.

No accuracy winner is established: the preliminary photogrammetry millimeter comparisons and direct-FLAME image-landmark agreement measure different things. They cannot be subtracted to claim an improvement. ARKit is an incumbent comparator, not error-free ground truth. A faster internal fitting timer is also not a measured time to a usable recommendation.

What is direct-photo-to-FLAME fitting?

FLAME is a reusable statistical 3D head model whose shape can be adjusted through numerical parameters. In direct-photo fitting, the software adjusts those shape parameters and the camera pose to align model landmarks with landmarks detected in photographs. This experiment fits one shared neutral head shape across nine training photos; it does not first build a photogrammetric surface.

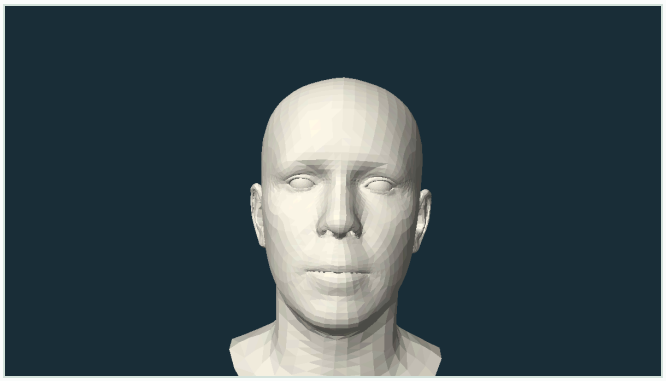
Below, the saved fitted head is rendered with the camera pose, focal length, and image framing estimated for the photograph beside it. The head orientation therefore follows the photo rather than using an arbitrary front or side view.

Front view

Photo capture · fitting input



Fitted FLAME head · matched saved pose

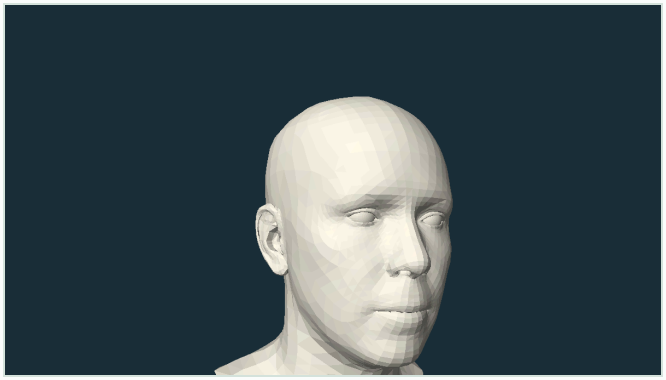


Turned view

Photo capture · fitting input



Same fitted head · matched saved pose



Actual saved 100-component, prior-100 direct-photo fit from the private experiment viewer. Both displayed photos were used for fitting; these are explanatory comparisons, not held-out validation. The same fitted identity is shown in two poses. Model surfaces are untextured; scalp, ears, and neck have no direct landmark constraints in this baseline. Physical scale remains uncalibrated, and visual resemblance does not establish millimeter accuracy. No new shape fitting or manual pose matching was performed for these figures. Model credit: FLAME 2023 Open; Li et al., *Learning a model of facial shape and expression from 4D scans*, 2017; official FLAME project.

Why compare this with photogrammetry? Direct fitting restricts the result to shapes the statistical model can represent, which could simplify or speed up processing but miss individual details. Photogrammetry instead reconstructs a surface from overlapping views. The research will measure whether any extra detail changes facial measurements and useful recommendations enough to justify the added workflow cost.

Healthcare use cases and the decision being supported

Candidate settings include recommending available respirator models and sizes to healthcare personnel before formal fit testing, supporting onboarding or periodic selection campaigns, and helping a central occupational-health team prepare shortlists for distributed staff. A fast assisted workflow may matter during a busy clinic session; a slower remote analysis may be acceptable before a scheduled appointment. Device access, connectivity, staff time, supported products, and the cost of unsuccessful first choices differ by setting.

The study will evaluate selection support within the intended workflow; geometry or a model score will not substitute for testing the worn respirator. Fit outcomes, candidate availability, and any assistance will be recorded consistently across methods. Initial operating assumptions and acceptable turnaround times will be agreed with prospective users before the locked comparison.

How much does the first approach cost versus the second?

There is no measured per-person answer yet. Instrument both pipelines from capture start through a usable ranked result. Record active device/CPU/GPU time, cold versus warm starts, upload/queue delay, scale entry, manual review, retries, staff assistance, and failed attempts. Price measured resource use with dated deployment-specific rates and include storage, transfer, model licensing, device provisioning, and support where applicable.

```
cost per usable result = total workflow cost, including failures and retries / usable
results
workflow cost = compute + storage/transfer + staff time + allocated
device/license/support costs
```

Show marginal operating cost and fully allocated cost separately. Account for research/development spending separately from recurring delivery cost; do not divide the entire research grant by a pilot's sample size and label it the production price. For ARKit, report both existing-device and purchased/shared-device scenarios, with an explicit amortization period and workload.

> **Illustrative compute calculation—not measured timings or deployment prices**
>
> At the proposal's assumed $2 per compute-hour, a hypothetical 120-second photogrammetry job costs about **$0.067** in active compute, versus **$0.0056** for a hypothetical 10-second direct-photo job: approximately **$0.061 extra per attempt**. At 1,000 attempts that is about $66.67 versus $5.56 in active compute alone.
>
> These durations are invented to show the calculation; they are not estimates derived from the current 6.96-second FLAME run or a forecast for photogrammetry. Billing granularity, idle capacity, different hardware rates, retries, and other workflow costs can change the result. For comparison, one extra minute of staff time at an assumed $40/hour costs about $0.67—so a small compute-price difference may be outweighed by capture or assistance time.

How much better are the results—and is the difference substantial?

Use paired captures from the geometry cohort and separately acquired physical references. Lock each method, its scale procedure, smoothing/fit settings, quality gates, and analysis before opening the held-out set. Match devices/lighting where feasible and document method-specific capture requirements. Benchmark cold/warm operation and median/95th-percentile total time on specified hardware rather than compare unmatched research runs.

Evaluate three levels of benefit:

1. **Geometry:** paired changes in bias, mean/tail absolute millimeter error, feature repeatability, and failure rate. Include nose/cheek/chin features needed by each style; do not infer local accuracy from a smoother mesh.
2. **Decisions:** feed each measurement source into the same frozen recommender and feature representation. Measure top-choice/top-three agreement and ranking relevance on comparable observed candidates. Evaluate new style-specific models as a separate factor, so a better feature set is not misattributed to a better capture method.
3. **Verified fit and operation:** measure first-choice quantitative fit-test pass rates, time/staff burden, and cost per usable recommendation and verified first-choice pass. A millimeter improvement may be marginal if it rarely changes a recommendation; a modest ranking improvement may matter at large staff volumes. Report uncertainty rather than declaring superiority from the point estimate alone.

Prespecify operationally meaningful differences with users of the workflow before the untouched comparison. If two methods have similar fit outcomes within the available uncertainty, the cheaper/faster usable method may be preferred, or the study may be inconclusive. The pilot is not a powered equivalence or non-inferiority trial. Report cost/accuracy/latency tradeoffs and supported conditions instead of forcing a universal winner.

```
incremental cost per extra first-choice pass = (cost per person, method A − method B)
/ (pass rate A − pass rate B)
```

Hypothetical interpretation: if photogrammetry costs $1 more per person and improves first-choice pass rate by one percentage point, the incremental cost is $100 per additional pass; a ten-percentage-point gain gives $10 per additional pass. These are invented effects, not project results or adoption thresholds. Use the formula only for a positive benefit and interpret it with paired uncertainty and total workflow costs. Fit passes are not infection-prevention outcomes.

Keep the comparison within the existing study scope

The geometry cohort will provide the main three-way paired benchmark, with ARKit availability reported rather than assumed for everyone. In the prospective pilot, direct-FLAME recommendations can be scored alongside photogrammetry and ARKit without automatically adding a third list of masks to test. Retain the current up-to-seven-option union. For cross-method first-choice pass comparisons, use a prespecified common observed candidate set and report untested recommendations as missing; do not silently impute their outcomes. If enough method-specific top choices are untested, a definitive comparative fit study requires a separately costed extension.

Feasibility benchmarking and bounded direct-fit evaluation sit within A/B/C, usability and deployment economics within E, and measured fit outcomes within D. The $196,800 planning envelope is unchanged. A materially larger comparison or fit-test burden would be explicitly re-scoped. Deliver a reproducible benchmark, deployment-cost worksheet, scenario analysis for healthcare workflows, and a proceed/narrow/stop recommendation.

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4. Research questions and comparisons

1. **Capture and scale:** how repeatable are RGB reconstructions, anatomical landmarks, and self-measured scale arcs across people, sessions, lighting, and devices?

2. **Smoothing:** does 3D Gaussian filtering improve independent physical measurement agreement compared with raw photogrammetry and zero-filter remeshing, and where does it remove useful nasal geometry?
3. **Measurement to ranking:** which compact spans, surface paths, and local nasal features provide incremental recommendation value for filtering facepiece respirators (FFRs) and quarter-mask elastomerics after realistic measurement uncertainty is propagated?
4. **Human usefulness:** does the first frozen camera-based recommendation improve quantitative fit-test pass rate relative to a participant's own initial choice?
5. **Cost-effectiveness:** how do photogrammetry, direct-photo-to-FLAME fitting, and iPhone-only ARKit compare in physical accuracy, time to a usable recommendation, and cost per usable result or verified first-choice fit-test pass? Are any gains operationally substantial rather than marginal?

Compare raw photogrammetry, zero-filter remeshing, Gaussian candidates, and the incumbent ARKit workflow where available. Independent physical references anchor accuracy. The same photos used for reconstruction or learned correspondences will not serve as the sole validation reference. Synthetic/printed faces remain a bounded diagnostic and stress-testing tool; they do not inherit human respirator-fit labels.

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5. Work plan and deliverables

A · Protocol, physical references, and repeatability · active months 1–3

Standardize landmark and route definitions, tape/caliper procedures, capture quality checks, and scale instructions. Use measured prints and repeat captures to test scale recovery, remeshing, and boundary effects. Freeze an initial protocol before new-person evaluation. Obtain appropriate ethics review and consent before funded recruitment.

Deliver: versioned capture/reference protocol; repeatability and scale-sensitivity analysis; 20-person geometry pilot; feasible-method and continuation recommendation.

B · Camera methods and local features · active months 2–6

Benchmark the existing photogrammetry/convolution pipeline against direct-photo-to-FLAME fitting and ARKit, including accuracy, full turnaround time, and workflow cost, before committing to a replacement estimator. Compare smoothing widths and zero-filter controls, landmark uncertainty, and model abstention. Test compact local geometry and feature-by-style effects against the current perimeter baseline on development data, following the bounded comparison above. Use limited synthetic rendering or printed references where they resolve a measured failure, rather than assuming synthetic training is necessary.

Deliver: reproducible baselines and ablations, fixed evaluation definitions, runtime and capture-burden measurements, and a documented selection of the method to validate. The existing 61-photo research workflow is not assumed to be an acceptable final user experience.

C · Human geometry benchmark · active months 2–8

Recruit approximately 120 adults: 20 pilot plus 40 additional development participants, and 60 untouched test participants. Assign every image, session, and derivative of one person to the same partition. Seek variation in facial shape, skin tone, age, gender, facial hair, and device access; report the resulting coverage without promising precise subgroup estimates from small cells.

Collect repeated unmarked RGB captures, physical tape/caliper measurements, independent scan references where available, and paired ARKit captures on supported hardware. Use a second rater on a prespecified subset. Reference scans must be separately acquired and independently scaled. Evaluate roughly six Android devices across camera capabilities, desktop/laptop webcams, and an iPhone reference; reserve at least one device model from development.

Freeze preprocessing, scale rules, landmarks, smoothing, capture-quality thresholds, and analysis before opening the 60-person test set. Report signed bias, mean and 95th-percentile absolute error in mm, relative error, repeatability, Bland–Altman agreement, and capture/abstention rates. Use participant-level resampling for repeated observations. Any scale-aligned shape-only diagnostic is separate from absolute millimeter accuracy.

Deliver: a locked measurement benchmark with participant flow, device/shape diagnostics, reference uncertainty, and proceed/narrow/stop recommendation.

D · Prospective human fit-testing pilot · active months 8–11

Conditional on the geometry gate, recruit approximately 60 separate adults who have not previously used MasqFit. Freeze the recommender before outcomes are collected. Each person first chooses a respirator from a predefined locally available set, then receives three camera-based recommendations. Where paired ARKit capture is available, also generate its three recommendations; test the union, up to seven distinct model/size combinations.

Randomize test order, standardize donning assistance, and blind the operator to recommendation source where practicable. Select an appropriate quantitative protocol and instrument configuration for the included respirator class before collection. Record continuous fit factors and protocol-defined pass/fail. Keep seal checks, qualitative tests, and quantitative outcomes distinct. Base the main pilot on a bounded FFR set; analyze any exploratory elastomeric results separately.

Primary: paired difference in quantitative pass rate between the first camera recommendation and initial choice. **Secondary:** at least one passing model among the camera top three, paired camera/ARKit results, test burden, and completion. Prespecify handling of technical failures, abstentions, missing outcomes, and withdrawals; include all enrolled people in the operational denominator.

This is a feasibility and effect-estimation pilot. For context, a single proportion near 50% at n=60 has an approximate 95% margin of error of 13 percentage points; paired-difference precision depends on discordant outcomes. It is not powered to establish non-inferiority, infection reduction, or subgroup efficacy.

Deliver: preregistered protocol, participant-flow report, paired effect estimates with uncertainty, and a larger-study sample-size plan.

E · Android/web beta and dissemination · active months 5–12

Translate the selected approach into guided capture, scale entry, quality feedback, and recommendation integration. Establish acceptable runtime, supported conditions, and user burden from the pilot before release. Produce a deployment-cost worksheet and healthcare-workflow scenario analysis for the three methods, separating marginal operating cost from fully allocated cost. Server-side processing is an option; this proposal does not assume photogrammetry runs on every phone or keeps all images on-device.

Deliver: a research beta within validated conditions, public technical report, reusable code and evaluation tooling, and documented negative or inconclusive findings.

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6. Milestones and funding decisions

These are provisional engineering targets. Reference uncertainty and recommender sensitivity will inform final tolerances during development, before the untouched test set is examined.

Decision	Evidence and target	Action if unmet
G1 · Feasibility Month 3	20-person pilot; repeatable landmark/reference procedures; full calibration-arc coverage; plausible independent-span accuracy and capture burden.	Refine protocol or scale aid; narrow scope or stop. Release the feasibility findings before seeking continuation.
G2 · Measurement Months 6–8	For critical spans, target ≤2 mm MAE and ≤5 mm 95th-percentile absolute error. For each aggregate path, target ≤5 mm MAE. Report each feature, scale bias, tails, and ranking sensitivity.	Retain research-only status or limit supported conditions. Do not pass a method on its mean while concealing tails or failed captures.
G3 · Usability Months 8–10	Target ≥90% capture completion within two attempts in supported conditions; record time, compute cost, and the full failure/abstention denominator.	Improve guidance, simplify capture, or require assistance. Do not claim untested device support.
G4 · Fit and release Month 12	60-person fit pilot; frozen predictions; paired estimates and uncertainty; complete participant accounting.	Publish inconclusive/negative findings and redesign or size a follow-up. Release only within supported scope.

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7. Budget and duration

The budget supports future protocol development, independent validation, integration, and dissemination, building on the existing photogrammetry and convolution experiments.

Proposed budget · USD · 180 active PI days at $600/day ($75/hour)

Workstream	PI days	PI effort	Other direct	Total
A · Reference protocol and feasibility	25	$15,000	$7,000	$22,000
B · Camera methods and local features	40	$24,000	$8,000	$32,000
C · Human geometry benchmark	35	$21,000	$15,000	$36,000
D · Prospective fit pilot	40	$24,000	$20,000	$44,000
E · Android/web beta and dissemination	40	$24,000	$2,000	$26,000
Direct research subtotal	180	$108,000	$52,000	$160,000
Research contingency · proposed, to be allocated to approved research costs				$20,000
Research funding subtotal				$180,000
Separate paid parental-leave allowance				$16,800
Total proposed request				$196,800

Resource assumptions

- **A · $7,000:** printing/measurement supplies $1,500; device pool $2,000; specialist anthropometry, statistical, and ethics-review allowance $3,500.
- **B · $8,000:** compute $4,000 (2,000 provisioned hours at an assumed $2/hour); model/asset rights allowance $2,000; secure storage and tooling $2,000. This is a planning ceiling, not a measured production inference price.
- **C · $15,000:** 120 participants × $50 = $6,000; measurement assistant 80 hours × $50 = $4,000; reference scan access $3,000; travel/retention $2,000.
- **D · $20,000:** 60 participants × $100 = $6,000; fit tester 160 hours × $40 = $6,400; up to 420 respirators × $5 = $2,100; instrument access/probes/consumables $4,000; logistics $1,500.
- **E · $2,000:** hosting/distribution $1,000; accessibility/usability review $1,000.

All rates are planning assumptions, not vendor quotations. Existing printer ownership is in-kind; no free institutional staff, instrument access, or specialist services are assumed. Additional retesting, recruitment difficulty, and reference costs may require scope adjustment or approved contingency use. No separate full-time engineer is budgeted. Within B, ten PI days ($6,000) are reserved for bounded feature comparisons, included in the total.

Initial commitment and conditional continuation

Stage	Allocation	Amount
1 · Three active months	A $22,000 + initial B $16,000 + C pilot $6,000 + research contingency $6,000 = $50,000; separate proposed leave $16,800.	$66,800
2 · Months 4–12, conditional	Remaining B $16,000 + C $30,000 + D $44,000 + E $26,000 + contingency $14,000.	$130,000
Same full scope, partitioned once		$196,800

The $50,000 research feasibility phase is a standalone option if leave is not included or the award begins after leave. It does not promise full deployment and the prospective trial. Even an unsuccessful result yields a reusable protocol, benchmark, and failure analysis.

Parental leave and calendar

The current plan includes eight weeks of parental leave around the expected December 17, 2026 birth, provisionally mid-December through mid-February. Compensation is calculated as 8 weeks × 28 funded hours/week × $75/hour = $16,800, separate from 1,440 active research hours. The milestone clock and routine PI reporting pause during leave; participant-facing obligations must be completed or assigned to an agreed qualified person beforehand. No additional coverage hire is assumed.

Twelve active months plus the pause imply approximately fourteen calendar months. Actual dates depend on award timing and the birth. If the award starts after leave, the schedule and allowance will be rebased with Blueprint rather than assuming retroactive payment. Recruitment/review delays may extend elapsed delivery without automatically increasing the award.

Overhead

No institutional overhead is budgeted. The $20,000 contingency is not an overhead charge; its allocation and treatment require agreement before award. Blueprint's published guidance caps indirect costs at 10% of direct costs and prefers projects of twelve months or less, with a twenty-four-month maximum. The staged initial scope allows an early decision; the full calendar including leave would need agreement. Source 2.

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8. Public outputs, participation, and team

Lead: Edderic Ugaddan, Breathesafe LLC, responsible for research design, modeling, engineering, analysis, and reporting. Paid measurement and fit-testing assistants and limited specialist review are included as allowances; no named external collaborator or institution is committed by this proposal.

Community contributors and practical experience

Community fit testers and mask-selection advocates bring practical experience that informs MasqFit's development and the research questions in this proposal:

- **Wayne — community fit tester.** Brought community fit testing to 12+ cities in the US and counting, personally tested 200+ people and trained dozens of community fit testers.
- **Mish — initial mask recommendations and community fit tester, MaskTogetherColorado.** Mish's *Masks for Kids* chart helps families navigate options across smaller face sizes.
- **Lisa Foreman — community fit tester.** Popularized the Readimask Dental Nose Hack.
- **Gerard Hughes — breathability and fit-testing data.** His mask fit testing, reviews, hacks, and experiments explore the balance of fit, filtration, comfort, and utility in respirator-grade masks.
- **Amanda Abbott — community fit tester.** Created the PDX Respirator Repository spreadsheet, with data on more than 60 different elastomeric respirators.

Planned paid support: trained measurement assistance, a quantitative fit tester, and targeted anthropometry, statistical, ethics, and usability review within the stated budget allowances. These roles are to be arranged; the existing contributions above do not imply new commitments to this funded project.

Public outputs and participation

Publish project-owned code with explicit compatible open-source licensing, reproducible settings, benchmark methods, and aggregate results, including negative findings. Model and asset license review precedes redistribution. Synthetic geometry or consented data may be released under appropriate terms; participant photographs, identifiable meshes, restricted assets, and every trained weight are not automatically public deliverables.

Recruit adults initially. Obtain appropriate ethics review, consent, and separately stated permissions for reconstruction, model development, and public sharing. Use pseudonymous records, access-controlled encrypted storage, a retention/deletion policy, and clear withdrawal limits. Pediatric validation would require a separately scoped study.

As an LLC, Breathesafe would agree terms supporting Blueprint's charitable purpose, broad public access, and incidental private benefit. Existing work and any third-party restrictions would be identified before contracting. The proposal does not imply that Blueprint has approved eligibility, budget terms, or funding.

During active months, provide brief progress/spending updates and milestone reviews; submit a final methods/results report and a continuation decision at the end of feasibility. Reporting cadence will respect the agreed leave period and Blueprint's grant workflow.

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9. References and evidence

1. Blueprint Biosecurity: Respiratory Protection. Program context: barriers to effective respirator use.
2. Blueprint Biosecurity: Funding Opportunities. Public-output expectations, non-tax-exempt award considerations, timeline guidance, and overhead policy; consulted November 9, 2026. This invitation is exploratory and outside a listed PPE RFP.
3. CDC/NIOSH: Fit Testing. Selection recommendations complement testing of the worn respirator.
4. Breathesafe source and MasqFit source; MasqFit on the App Store.

Recommender results: live dashboard retrieved November 9, 2026; model 20261108020017. Metric definition checked against the project's user-grouped top-k evaluation implementation. The local evidence register records the retrieved figures and source-code hash.

Internal evidence snapshot: November 9, 2026. Width candidates v4; face-length check v1; arc calibration v1; selected upper-neck Gaussian surface; all-61-view support review. Exact report hashes and arithmetic are preserved in the local evidence register. These feasibility results have not been independently replicated.

Prepared for discussion with Victoria Slaughter and Nickia, Blueprint Biosecurity. No funding commitment or review timetable is assumed. This standalone HTML embeds its figures for online and offline review.